

# PAPER The Centrepost Life-Limiter: Fast-Fluence, dpa, and Shielding Setting the Cartridge-Replacement Cadence for a Compact Spherical-Tokamak Breeder

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## Abstract

The inboard centrepost is the defining vulnerability of a normal-conductor spherical tokamak: it sits closest to the 14.06 MeV fusion source and cannot be thickly shielded without growing the machine or displacing the breeding blanket. We do not argue this away; we quantify it and design for it. We solve the steady linear Boltzmann transport equation for the Hyperion breeder ( $R_0 = 1.20$  m, aspect ratio  $A = 2.5$ ,  $B_0 = 8$  T,  $I_p = 9.66$  MA,  $P_{\text{fus}} = 85.04$  MW, negative triangularity  $\delta = -0.30$ , elongation  $\kappa = 2.0$ , centrepost peak field 16.84 T) with Monte-Carlo neutron transport on ENDF/B-VIII.0, scoring the fast-neutron fluence ( $E > 0.1$  MeV) and the displacement damage at the centrepost REBCO tape behind a tungsten-carbide (WC) shield of variable thickness, and we fold the tallied spectrum against a Norgett–Robinson–Torrens (NRT) damage function corrected by an athermal-recombination (arc-dpa) efficiency. A 45 cm inboard standoff attenuates the centrepost fast fluence by a factor of 169, from  $2.93 \times 10^{25} \text{ n m}^{-2} \text{ fpy}^{-1}$  bare to  $1.73 \times 10^{23} \text{ n m}^{-2} \text{ fpy}^{-1}$  at full WC, with an effective removal coefficient that hardens from  $0.070 \text{ cm}^{-1}$  near the source to  $0.141 \text{ cm}^{-1}$  in the clean deep spectrum. At the design shield the reference structural damage rate is  $18.85 \text{ fpy}^{-1}$  and, at the REBCO fast-fluence damage ceiling  $\Phi_{\text{allow}} \approx 3.9 \times 10^{22} \text{ n m}^{-2}$ , the cartridge life is 0.22 fpy; a thin 5 cm shield gives 0.0048 fpy and a thick 40 cm shield 0.194 fpy, so shielding trades directly and monotonically against inboard breeding coverage (net 1-D TBR falls from 1.035 to 0.923). The centrepost-life ceiling caps the admissible plasma current to [8.78, 9.66] MA; the mid-current point extends life to 0.496 fpy ( $\times 2.25$ ). The design consequence is a maintenance architecture — a demountable, scheduled-swap cartridge at  $\sim 4.5\text{--}4.9 \text{ swaps fpy}^{-1}$  — not a materials miracle. Every headline quantity is a frozen anchor in the Kronos de-risking register.

**Keywords:** centrepost; fast neutron fluence; displacement damage; dpa; tungsten-carbide shield; REBCO coated conductor; scheduled-replacement cartridge; spherical-tokamak breeder

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## 1 Introduction

### 1.1 The centrepost problem

A spherical tokamak (ST) confines a large plasma current in a small volume, which is exactly what makes it an efficient 14 MeV neutron source and tritium breeder [1, 2], and, at the same time, what forces its hardest engineering compromise. To reach aspect ratio  $A = 2.5$  at major radius  $R_0 = 1.20$  m the inboard toroidal-field return — the *centrepost* — must be slender, and it therefore sits only tens of centimetres from the plasma with almost no room for shielding. Every high-field ST power-plant study confronts the same fact: the centrepost is the most neutron-exposed structural and magnetic element in the machine, and no shield that fits inside the geometry can protect it to a multi-decade life. This is the price of compactness, and it is best paid openly.

The response taken here follows the demountable-magnet philosophy of ARC [3] to its logical conclusion: rather than claiming a lifetime centrepost, we treat it as a *consumable*, *scheduled-replacement cartridge* and design the machine for its periodic removal. The engineering question then changes from an intractable materials problem (“how do we make a magnet survive  $10^{27}$  n m<sup>-2</sup>?”) to a tractable maintenance-interval problem (“how often must the cartridge be swapped, and can the plant tolerate that cadence?”). Answering it requires three quantities computed to nuclear-data fidelity: the fast-neutron fluence at the conductor, the displacement damage in the surrounding structure, and the way both scale with an inboard shield that is itself in competition with the breeding blanket. This paper computes all three, and converts them into a replacement cadence.

### 1.2 Prior art

The neutron shielding of superconducting cores in compact tokamaks has an established literature. Windsor and co-workers mapped the neutron and gamma flux through tungsten- and tungsten-boride-based shields protecting the superconducting centre column of an ST power plant, and quantified the heat deposition and damage that a practical shield must suppress [7, 8, 9]. The ARC line established the compact high-field, demountable-magnet reactor concept and the associated inboard-build pressure [3, 4]. Systems codes such as PROCESS carry parametric centrepost and shield models across the ST design space [5, 6]. On the materials side, the displacement-damage framework rests on the Norgett–Robinson–Torrens (NRT) standard [10], its Kinchin–Pease [11] and Lindhard [12] antecedents, and the modern athermal-recombination (arc-dpa) correction of Nordlund *et al.* [13, 14] with effective threshold and efficiency evaluations by Konobeyev *et al.* [15]; the fusion-materials context is set by Zinkle [23, 24], Gilbert [25], Stork [26] and the DEMO programme [27]. The specific vulnerability of REBCO coated conductors to fast-neutron irradiation — a critical-current degradation onset in the  $10^{22}$ – $10^{23}$  n m<sup>-2</sup> range — has been measured by Fischer, Prokopec, Eisterer and co-workers [19, 20] and underpins the high-field no-insulation (NI) REBCO magnets that make the compact ST possible [21, 22].

### 1.3 Contribution

What the prior art leaves open for a *breeder* is the coupled ledger: the same 45 cm of inboard radius is contested between the shield that buys centrepost life and the mantle that breeds tritium, and the correct design point is the one that respects both. This paper contributes (i) a formal statement of the transport-and-damage problem with the attenuation, NRT/arc-dpa and damage-limited-life functionals written explicitly (§2); (ii) a high-statistics Monte-Carlo solution on ENDF/B-VIII.0 that resolves the centrepost fast-fluence attenuation to sub-percent tally error and extracts a depth-dependent removal coefficient (§3–4); (iii) the coverage-versus-protection optimisation cast as a constrained program whose Karush–Kuhn–Tucker (KKT) solution is the observed corner design (full WC standoff, outboard breeding); (iv) the map from damage ceiling to cartridge life to replacement cadence and fleet availability; and (v) the current-window coupling that ties the centrepost-life ceiling to the admissible plasma current. Every number traces to a frozen serial in the Kronos Physics De-Risking Register [33], cited inline (e.g. BR-L1-A13b). The result sits within a companion set: the consumable-centrepost neutronics and coil-integrity overview [34], the first-principles REBCO radiation-tolerance study that fixes the fluence ceiling [35], the operating-window synthesis [36], the tritium-breeding fleet case [37], the first-wall damage-and-lifetime study [38], the prognostics and remaining-useful-life framework [39], and the swap-logistics and fleet-RAMI analyses [40, 41] that consume the cadence derived here.

## 2 Governing equations and the formal problem

### 2.1 Neutron transport

The centrepost fluence is the solution of the steady-state, fixed-source linear Boltzmann transport equation for the angular neutron flux  $\psi(\mathbf{r}, \boldsymbol{\Omega}, E)$ ,

$$\boldsymbol{\Omega} \cdot \nabla \psi + \Sigma_t(\mathbf{r}, E) \psi = \int_0^\infty dE' \oint_{4\pi} d\boldsymbol{\Omega}' \Sigma_s(\mathbf{r}, E' \rightarrow E, \boldsymbol{\Omega}' \cdot \boldsymbol{\Omega}) \psi(\mathbf{r}, \boldsymbol{\Omega}', E') + \frac{S(\mathbf{r}, E)}{4\pi}, \quad (1)$$

with  $\Sigma_t$  and  $\Sigma_s$  the macroscopic total and differential-scattering cross-sections from ENDF/B-VIII.0, and  $S$  the fusion source. For a D–T plasma the source is a near-monoenergetic 14.06 MeV emission,

$$S(\mathbf{r}, E) = S_n f_{\text{src}}(\mathbf{r}) \chi_{\text{DT}}(E), \quad S_n = \frac{P_{\text{fus}}}{\mathcal{E}_{\text{DT}}} = \frac{85.0 \times 10^6 \text{ W}}{17.6 \text{ MeV} \times 1.602 \times 10^{-19} \text{ J eV}^{-1}} = 3.016 \times 10^{19} \text{ s}^{-1}, \quad (2)$$

where  $\chi_{\text{DT}}$  is the Gaussian-broadened D–T line and  $f_{\text{src}}$  its spatial distribution (a toroidal ring at  $R_0$ ). The scalar flux is  $\phi(\mathbf{r}, E) = \oint_{4\pi} \psi d\boldsymbol{\Omega}$ , and the neutron wall loading that anchors the reference damage rate is  $\Gamma_w = 1.885 \text{ MW m}^{-2}$  (serial BR-L1-A13).

### 2.2 The fast-fluence functional and the attenuation model

The quantity that limits the REBCO conductor is the accumulated fast fluence above the damage threshold  $E_c = 0.1 \text{ MeV}$  at the peak-tape monitor cell  $\mathbf{r}_{\text{cp}}$ ,

$$\dot{\Phi}_{\text{fast}}(\mathbf{r}_{\text{cp}}) = \int_{E_c}^\infty \phi(\mathbf{r}_{\text{cp}}, E) dE, \quad \Phi_{\text{fast}}(t) = \int_0^t \dot{\Phi}_{\text{fast}} dt', \quad (3)$$

reported per full-power year (fpy). Along the inboard radius the transmitted fast fluence through a shield of thickness  $x$  obeys the classical removal-plus-buildup form

$$\dot{\Phi}_{\text{fast}}(x) = \dot{\Phi}_0 B(x) \exp\left[-\int_0^x \Sigma_r(x') dx'\right], \quad (4)$$

with  $\Sigma_r$  the macroscopic removal cross-section for fast neutrons in the shield material and  $B(x) \geq 1$  the buildup factor accounting for in-scattered neutrons that re-enter the fast group. Equation (4) defines an *effective* depth-dependent removal coefficient,

$$\Sigma_r^{\text{eff}}(x) \equiv -\frac{d}{dx} \ln \dot{\Phi}_{\text{fast}}(x), \quad (5)$$

which we extract directly from the Monte-Carlo tally (§4) and which rises with depth as the buildup of degraded neutrons is progressively removed and the surviving spectrum hardens toward the clean 14.06-MeV attenuation length.

### 2.3 The displacement-damage functional

The structural damage that governs the steel liner, first wall and centrepost jacket is the displacement rate. A neutron of energy  $E$  transfers recoil energy  $T$  to a lattice atom with cross-section  $d\sigma/dT$ ; only the nuclear-stopping part of  $T$  produces displacements, given by the Lindhard partition

$$T_{\text{dam}}(T) = \frac{T}{1 + k g(\varepsilon)}, \quad g(\varepsilon) = 3.4008 \varepsilon^{1/6} + 0.40244 \varepsilon^{3/4} + \varepsilon, \quad (6)$$

with  $\varepsilon$  the reduced recoil energy and  $k$  the Lindhard electronic-stopping constant. The Norgett–Robinson–Torrens number of stable Frenkel pairs per primary knock-on atom (PKA) is [10]

$$\nu_{\text{NRT}}(T_{\text{dam}}) = \begin{cases} 0, & T_{\text{dam}} < E_d, \\ 1, & E_d \leq T_{\text{dam}} < \frac{2E_d}{0.8}, \\ \frac{0.8 T_{\text{dam}}}{2E_d}, & T_{\text{dam}} \geq \frac{2E_d}{0.8}, \end{cases} \quad (7)$$

with  $E_d$  the effective threshold displacement energy. The NRT displacement rate then folds the flux against the damage cross-section,

$$R_{\text{dpa}}^{\text{NRT}} = \frac{0.8}{2E_d} \int_0^\infty \phi(E) \left[ \int_{E_d}^{T_{\text{max}}(E)} \frac{d\sigma}{dT} T_{\text{dam}}(T) dT \right] dE \equiv \int_0^\infty \phi(E) \sigma_D(E) dE, \quad (8)$$

where  $\sigma_D(E)$  is the NJOY-processed damage-energy cross-section (reaction MT= 444). Molecular-dynamics cascade evidence shows the NRT number overestimates surviving damage because of athermal in-cascade recombination; the arc-dpa model [13, 14] multiplies  $\nu_{\text{NRT}}$  by an efficiency

$$\xi_{\text{arc}}(T_{\text{dam}}) = \frac{1 - c_{\text{arc}}}{(2E_d/0.8)^{b_{\text{arc}}}} T_{\text{dam}}^{b_{\text{arc}}} + c_{\text{arc}}, \quad \xi_{\text{arc}} \rightarrow c_{\text{arc}} \approx 0.3 \text{ as } T_{\text{dam}} \rightarrow \infty, \quad (9)$$

so that at the  $\mathcal{O}(10\text{--}30 \text{ keV})$  cascade energies characteristic of 14-MeV recoils in Fe and W (serial BR-L2-A24) the surviving-damage rate is  $R_{\text{dpa}}^{\text{arc}} \approx 0.3 R_{\text{dpa}}^{\text{NRT}}$ . We report the register anchor  $18.85 \text{ fpy}^{-1}$  on the NRT convention (BR-L1-A13; wall-loading-anchored value  $19.66 \text{ fpy}^{-1}$ , BR-SX-02) and carry the arc-dpa efficiency as the conservatism margin.

#### 2.4 Damage-limited cartridge life

The centrepost is retired when its accumulated fast fluence reaches the REBCO critical-current-degradation ceiling  $\Phi_{\text{allow}}$ . The cartridge life and replacement cadence are therefore

$$T_c(x) = \frac{\Phi_{\text{allow}}}{\dot{\Phi}_{\text{fast}}(x)}, \quad n_{\text{swap}}(x) = \frac{1}{T_c(x)} = \frac{\dot{\Phi}_{\text{fast}}(x)}{\Phi_{\text{allow}}}, \quad (10)$$

in fpy and swaps  $\text{fpy}^{-1}$  respectively. The ceiling is not a free parameter: first-principles DFT+ $U$  chain-oxygen-vacancy formation and YBCO displacement-cascade molecular dynamics place the onset at  $\Phi_{\text{allow}} \approx 3.9 \times 10^{22} \text{ n m}^{-2}$  for artificial-pinning-centre tape (un-doped break-even  $5.64 \times 10^{22} \text{ n m}^{-2}$ ; serials BR-SX-04 and companion [35]). We bracket the sensitivity with the commonly cited  $1 \times 10^{22}$  (conservative) and  $1 \times 10^{23} \text{ n m}^{-2}$  (optimistic) limits.

#### 2.5 Coverage versus protection as a constrained program

The inboard standoff of fixed thickness  $L = 45 \text{ cm}$  is partitioned between a WC shield of thickness  $x$  (inner, protecting the conductor) and a lithium breeding mantle of thickness  $L - x$  (outer, breeding tritium). Both the cartridge life and the net tritium-breeding ratio  $\Lambda(x)$  depend on  $x$ , and they pull in opposite directions: thicker WC raises  $T_c$  but starves the inboard mantle. The design point solves

$$x^* = \arg \max_{0 \leq x \leq L} T_c(x) \quad \text{s.t.} \quad \Lambda(x) \geq \Lambda_{\text{min}}, \quad (11)$$

with Lagrangian  $\mathcal{L}(x, \mu) = T_c(x) + \mu [\Lambda(x) - \Lambda_{\text{min}}]$  and stationarity

$$\frac{dT_c}{dx} + \mu \frac{d\Lambda}{dx} = 0, \quad \mu [\Lambda(x) - \Lambda_{\text{min}}] = 0, \quad \mu \geq 0. \quad (12)$$

Because  $T'_c(x) > 0$  and  $\Lambda'(x) < 0$  everywhere on  $[0, L]$  (§4), an interior optimum would require  $\mu > 0$  with the breeding constraint active. The 1-D radial calculation, however, returns  $\Lambda(x) < 1$  for *all*  $x$  (the inboard side cannot reach self-sufficiency), so the constraint  $\Lambda(x) \geq \Lambda_{\text{min}}$  with a self-sufficient  $\Lambda_{\text{min}}$  is infeasible inboard and the breeding target is instead met from the outboard blanket and multiplier (companions [37, 36]). With the inboard breeding requirement relaxed, (12) has its maximiser at the corner  $x^* = L$ : the full standoff is given to WC, and the inboard mantle breeds nothing. This corner solution is exactly the register's recommended operating shield (BR-L1-A13b).

#### 2.6 Current-window coupling

Because  $\dot{\Phi}_{\text{fast}} \propto S_n \propto P_{\text{fus}}$  and  $R_{\text{dpa}} \propto S_n$ , the centrepost-life ceiling is equivalent to a fusion-power ceiling and, through the confinement scaling, to a plasma-current ceiling. Writing the damage constraint  $R_{\text{dpa}}(I_p) \leq R_{\text{dpa}}^{\text{max}} = 18.85 \text{ fpy}^{-1}$  and the life floor  $T_c(I_p) \geq T_c^{\text{min}}$ , the admissible current window is (serial KX-L1-A21)

$$I_p \in [8.78 \text{ MA}, 9.66 \text{ MA}], \quad (13)$$

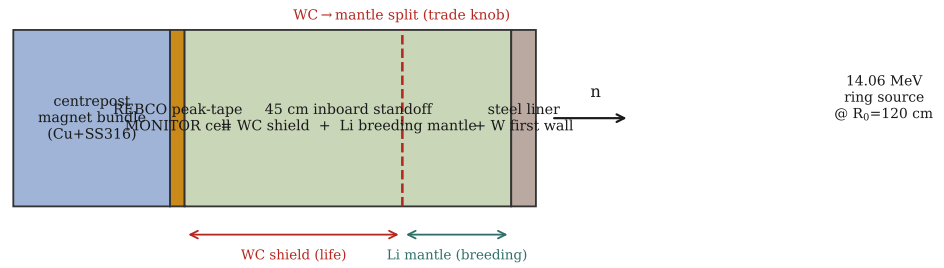
over  $q_{95} \in [2.7, 3.3]$ , with the tritium-startup floor non-binding. Reducing the operating point from the design current to the mid-current edge of (13) lowers the source strength and extends the cartridge life by the register factor  $T_c^{\text{mid}}/T_c^{\text{des}} = 0.496/0.22 = 2.25$  (serial BR-SX-16).

### 3 Computational methodology: the GPU-HPC campaign

#### 3.1 Monte-Carlo transport

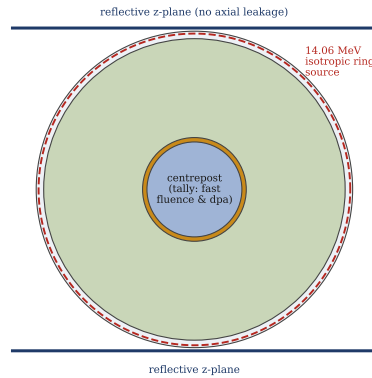
Equation (1) is solved by the open-source Monte-Carlo code OpenMC 0.15.3 [16] on the ENDF/B-VIII.0 continuous-energy library [17] (endfb80-lowtemp HDF5), continuous-energy and temperature-corrected. The frozen design point is transported in a one-dimensional radial infinite-cylinder constructive-solid-geometry (CSG) model: concentric  $z$ -cylinders about the machine axis with reflective  $z$ -planes (no axial leakage, clean radial transport), and a 14.06 MeV isotropic ring source at  $R_0 = 120$  cm uniform in azimuth and height (figures 1–2). The inboard radial build, from the axis outward, is: centrepost magnet bundle  $r = 0$ –21.6 cm (homogenised Cu 60 vol%+SS316 40 vol% “REBCO\_centrepost” proxy); the REBCO peak-tape monitor cell  $r = 21.6$ –23.6 cm (the fluence-tally location); a fixed 45 cm tradeable standoff  $r = 23.6$ –68.6 cm split as WC shield (inner) plus lithium breeding mantle (outer); then the steel liner and tungsten first wall; then the plasma. The trade knob is the shield thickness  $x = 45 - t_{\text{mantle}}$  for  $t_{\text{mantle}} \in \{0, 15, 30, 45\}$  cm (serial BR-L1-A13b).

(Schematic.) Inboard radial build and the shield / mantle trade knob



**Figure 1. (Schematic.)** Inboard radial build transported in the OpenMC model, from the machine axis outward: centrepost magnet bundle, REBCO peak-tape fluence-monitor cell, the 45 cm tradeable standoff split between WC shield (protection) and lithium breeding mantle (coverage), and the steel liner / tungsten first wall. The red dashed line is the shield→mantle trade knob of the constrained program (11). No data values are shown.

(Schematic.) 1-D radial CSG transport model (OpenMC)



**Figure 2. (Schematic.)** The 1-D radial infinite-cylinder CSG transport model: concentric  $z$ -cylinders about the machine axis with reflective  $z$ -planes (no axial leakage) and a 14.06 MeV isotropic ring source at  $R_0 = 120$  cm. The fast fluence and displacement damage are tallied at the centrepost monitor cell. The figure encodes structure only.

#### 3.2 Damage cross-sections and multiscale PKA modelling

The damage-energy cross-section  $\sigma_D(E)$  of equation (8) is generated with the NJOY nuclear-data processing system [18] (HEATR module, MT= 444) from the same ENDF/B-VIII.0 evaluations, and the NRT number of equation (7) is applied group-wise. To justify the arc-dpa efficiency of equation (9) at the cascade level, primary-knock-on spectra are computed with SPECTRA-PKA driven by the NJOY21 recoil matrices and propagated into molecular-dynamics displacement cascades in LAMMPS: 15-keV Fe–9Cr and tungsten cascades are run and analysed for surviving

Frenkel pairs by Wigner–Seitz occupancy (serial BR-L2-A24), with effective threshold and arc-dpa parameters cross-checked against the evaluations of Konobeyev, Fischer and Simakov [15, 14] and the multiscale radiation-damage chain of serial HX-20. The REBCO fast-fluence ceiling is set independently by DFT+ $U$  chain-oxygen-vacancy formation energies and YBCO displacement-cascade MD (companion [35]; serial BR-SX-04).

### 3.3 Scale, acceleration and reproducibility

Each of the four shield-split points was run at  $5 \times 10^7$  particles per batch over 50 active batches, for  $2.5 \times 10^9$  neutron histories per point, on the NVIDIA GPU HPC campaign (Lambda A100/H100/A10 fleet); the Monte-Carlo history transport and the batch-parallel tally reduction map onto the GPU fleet, while the NJOY processing, the LAMMPS cascade analysis and the trade-curve reduction run on the in-house HPC node so the two workloads never contend. The variance is small and machine-checkable: the net-TBR relative error is  $\sim 2 \times 10^{-5}$  and the centrepost fast-flux relative error is 0.02–0.24% across the split points (table 3). The transport model, source normalisation of equation (2), radial build and tally definitions are archived byte-for-byte and are regenerable from the cited serials; no result depends on proprietary data. Table 1 collects the codes, library, problem scale and verification metrics.

**Table 1.** Computational specification and reproducibility metrics for the centrepost transport-and-damage campaign. Serials refer to the Kronos de-risking register [33]. No compute-cost figures are reported.

| element            | specification                               | role                                        |
|--------------------|---------------------------------------------|---------------------------------------------|
| transport code     | OpenMC 0.15.3 [16]                          | fixed-source neutron transport, eq. (1)     |
| nuclear data       | ENDF/B-VIII.0 [17]                          | continuous-energy cross-sections            |
| damage processing  | NJOY (HEATR, MT= 444) [18]                  | damage-energy $\sigma_D(E)$ , eq. (8)       |
| PKA / cascade      | SPECTRA-PKA + NJOY21 $\rightarrow$ LAMMPS   | 15-keV Fe–9Cr, W cascades (BR-L2-A24)       |
| REBCO ceiling      | DFT+ $U$ + YBCO cascade MD                  | $\Phi_{\text{allow}}$ , eq. (10) (BR-SX-04) |
| geometry           | 1-D radial infinite-cylinder CSG            | reflective $z$ , ring source at $R_0$       |
| histories / point  | $5 \times 10^7 \times 50 = 2.5 \times 10^9$ | 4 shield-split points (BR-L1-A13b)          |
| hardware           | NVIDIA GPU HPC campaign                     | A100 / H100 / A10 fleet                     |
| TBR tally error    | $\sim 2 \times 10^{-5}$ (rel.)              | verification                                |
| cp fast-flux error | 0.02–0.24% (rel.)                           | verification                                |

## 4 Results

### 4.1 Design point and the reference damage rate

Table 2 lists the frozen design point transported here. At the neutron wall loading  $\Gamma_w = 1.885 \text{ MW m}^{-2}$  the reference NRT displacement rate is  $18.85 \text{ fpy}^{-1}$  (BR-L1-A13), the ceiling that the current window of equation (13) is built to respect. This is a first-wall-referenced structural rate; the fast fluence that actually retires the centrepost conductor is the attenuated quantity computed next.

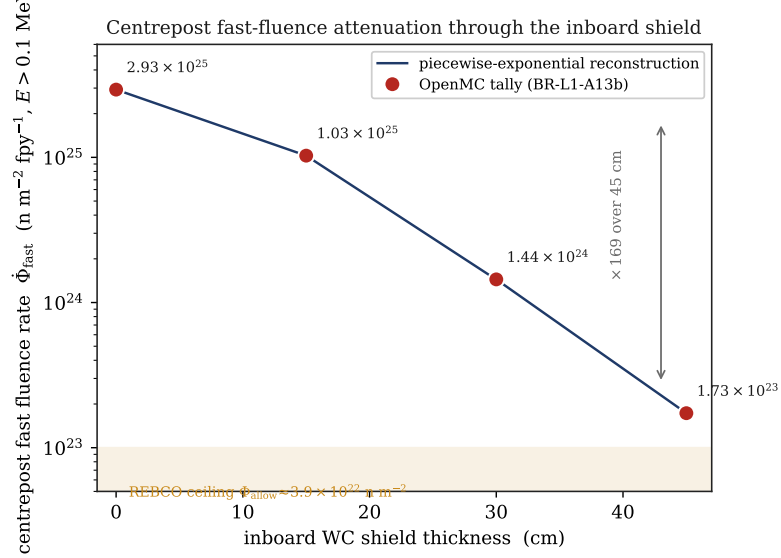
**Table 2.** Frozen Hyperion breeder design point (Tier-2 canonical anchors). Two distinct magnets are carried in the programme; this paper concerns the breeder centrepost only.

| quantity                   | symbol            | value                                 |                          |
|----------------------------|-------------------|---------------------------------------|--------------------------|
| major radius               | $R_0$             | 1.20 m                                |                          |
| aspect ratio               | $A$               | 2.5                                   | ( $a = 0.48 \text{ m}$ ) |
| elongation / triangularity | $\kappa / \delta$ | 2.0 / $-0.30$                         | negative- $\delta$       |
| on-axis field              | $B_0$             | 8 T                                   |                          |
| centrepost peak field      | $B_{\text{cp}}$   | 16.84 T                               | breeder magnet           |
| plasma current             | $I_p$             | 9.66 MA                               |                          |
| fusion power               | $P_{\text{fus}}$  | 85.04 MW                              | ( $Q = 3.076$ )          |
| source rate                | $S_n$             | $3.016 \times 10^{19} \text{ s}^{-1}$ | eq. (2)                  |
| neutron wall loading       | $\Gamma_w$        | $1.885 \text{ MW m}^{-2}$             | BR-L1-A13                |
| reference damage rate      | $R_{\text{dpa}}$  | $18.85 \text{ fpy}^{-1}$              | NRT, BR-L1-A13           |

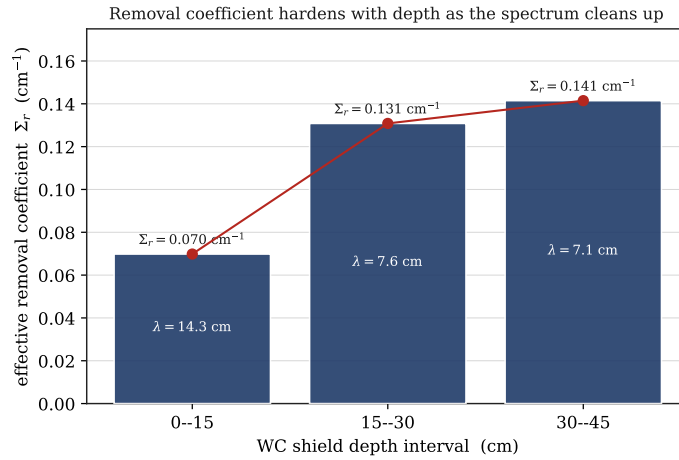
### 4.2 Fast-fluence attenuation and the removal coefficient

Figure 3 shows the centrepost fast fluence (equation 3) against WC shield thickness from the four high-statistics tallies. The fluence falls monotonically from  $2.928 \times 10^{25} \text{ n m}^{-2} \text{ fpy}^{-1}$  bare (full mantle, no shield) to  $1.729 \times 10^{23} \text{ n m}^{-2} \text{ fpy}^{-1}$  at full WC, a factor of 169 over the 45 cm standoff. The attenuation is not a single exponential: extracting the effective removal coefficient of equation (5) interval-by-interval (figure 4) gives  $\Sigma_r^{\text{eff}} = 0.070 \text{ cm}^{-1}$  over 0–15 cm,  $0.131 \text{ cm}^{-1}$  over

15–30 cm, and  $0.141 \text{ cm}^{-1}$  over 30–45 cm — an  $e$ -folding length that shortens from 14.3 to 7.1 cm with depth. This depth-hardening is the physical signature of buildup: near the source the fast group is continually re-populated by down-scattered 14-MeV neutrons ( $B(x) \gg 1$ ), while deep in the shield the surviving spectrum is clean and attenuates at the material removal length. The consequence for design is that the last centimetres of shield are the most effective per unit thickness.



**Figure 3.** Centrepost fast-neutron fluence rate ( $E > 0.1 \text{ MeV}$ ) versus inboard WC shield thickness, from four  $2.5 \times 10^9$ -history OpenMC tallies on ENDF/B-VIII.0 (serial BR-L1-A13b). The fluence falls by a factor of 169 over the 45 cm standoff; the gold band marks the REBCO damage-ceiling bracket ( $1\text{--}10 \times 10^{22} \text{ n m}^{-2}$ ).



**Figure 4.** Effective removal coefficient  $\Sigma_r^{\text{eff}}$  (equation 5) by shield-depth interval, extracted from the tallies of figure 3. The coefficient hardens from  $0.070$  to  $0.141 \text{ cm}^{-1}$  with depth as buildup is removed and the spectrum cleans up.

#### 4.3 Cartridge life and the damage ceiling

Converting fast fluence to life through equation (10) requires the REBCO ceiling  $\Phi_{\text{allow}}$ . Figure 5 plots the cartridge life against shield thickness at the conservative ( $1 \times 10^{22}$ ), REBCO ( $3.9 \times 10^{22}$ ) and optimistic ( $1 \times 10^{23} \text{ n m}^{-2}$ ) ceilings. At the corner design (full 45 cm WC,  $\dot{\Phi}_{\text{fast}} = 1.729 \times 10^{23} \text{ n m}^{-2} \text{ fpy}^{-1}$ ) the REBCO ceiling gives

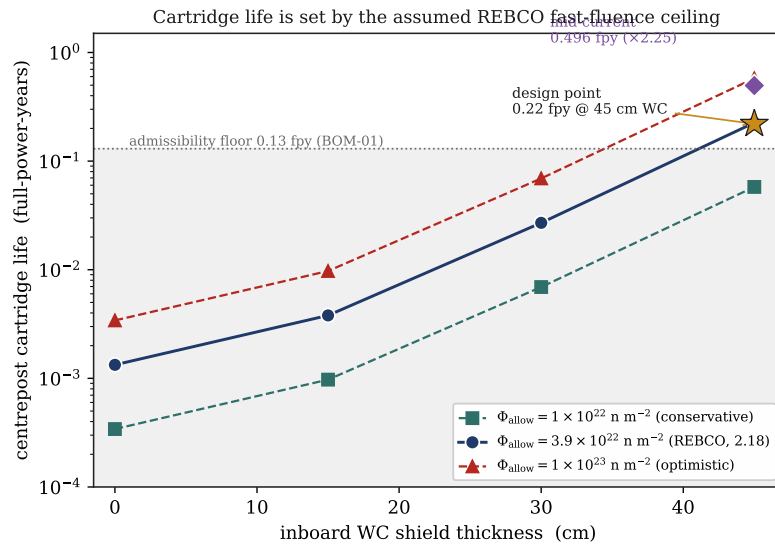
$$T_c = \frac{3.9 \times 10^{22} \text{ n m}^{-2}}{1.729 \times 10^{23} \text{ n m}^{-2} \text{ fpy}^{-1}} = 0.226 \text{ fpy}, \quad (14)$$

reproducing the register design anchor of 0.22 fpy (serial SC-BR-2: 30 cm-class WC gives 0.221 fpy). Thinner shields collapse the life: 5 cm yields 0.0048 fpy and the bare centrepost only

$\sim 0.003\text{--}0.004$  fpy ( $\sim 1.4$  full-power-days, inadmissible), while a thick 40 cm shield reaches 0.194 fpy. Every candidate at any point on the standoff falls below a lifetime-component threshold by two-to-three orders of magnitude: the compact-ST centrepost *cannot* be shielded to multi-fpy life within the available inboard radius, which is precisely why it is a scheduled-replacement cartridge and not a lifetime part. Table 3 gives the full trade.

**Table 3.** Centrepost fast-fluence / net-TBR shield trade (serial BR-L1-A13b; OpenMC 0.15.3, ENDF/B-VIII.0,  $2.5 \times 10^9$  histories/point). Life at three fast-fluence damage ceilings; the REBCO column uses  $\Phi_{\text{allow}} = 3.9 \times 10^{22} \text{ n m}^{-2}$  (companion [35]).

| WC<br>(cm) | Li mantle<br>(cm) | $\dot{\Phi}_{\text{fast}}$<br>( $\text{n m}^{-2}\text{fpy}^{-1}$ ) | net<br>TBR   | $T_c @ 10^{22}$<br>(fpy) | $T_c @ 3.9 \times 10^{22}$<br>(fpy) | $T_c @ 10^{23}$<br>(fpy) |
|------------|-------------------|--------------------------------------------------------------------|--------------|--------------------------|-------------------------------------|--------------------------|
| 0          | 45                | $2.928 \times 10^{25}$                                             | 1.035        | 0.00034                  | 0.0013                              | 0.0034                   |
| 15         | 30                | $1.027 \times 10^{25}$                                             | 1.022        | 0.00097                  | 0.0038                              | 0.0097                   |
| 30         | 15                | $1.444 \times 10^{24}$                                             | 0.997        | 0.0069                   | 0.0270                              | 0.0693                   |
| <b>45</b>  | <b>0</b>          | <b><math>1.729 \times 10^{23}</math></b>                           | <b>0.923</b> | <b>0.0578</b>            | <b>0.226</b>                        | <b>0.578</b>             |



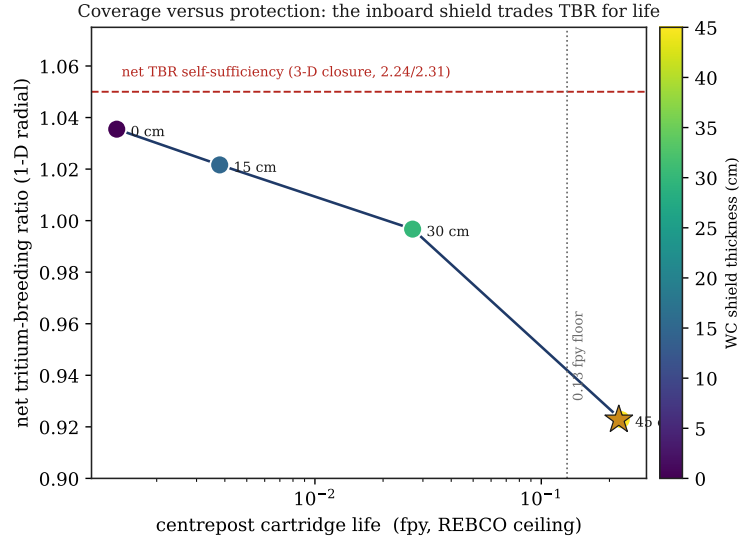
**Figure 5.** Centrepost cartridge life versus WC shield thickness at three fast-fluence damage ceilings. The design point (star) is the full 45 cm WC standoff at the REBCO ceiling, 0.22 fpy; the mid-current operating point extends it to 0.496 fpy. The grey band is the 0.13 fpy admissibility floor (serial BOM-01).

#### 4.4 Coverage versus protection

The same shield that buys centrepost life displaces the inboard breeding mantle. Figure 6 plots the net 1-D radial TBR against cartridge life as the shield–mantle split is swept, tracing the coverage-versus-protection Pareto front of the constrained program (11). Net TBR falls from 1.035 (bare) to 0.923 (full WC) while life rises by two-plus decades. Crucially the 1-D radial net TBR is below the self-sufficiency line  $\Lambda_{\text{min}} = 1.05$  at *every* split point — so, per the KKT analysis of §2.5, the inboard breeding constraint is infeasible and the optimiser sits at the corner  $x^* = L$ : the full standoff goes to WC and tritium self-sufficiency is recovered from the outboard blanket and neutron multiplier in the 3-D closure (net TBR  $\geq 1.05$ ; companions [37, 36]). The inboard shield is thus not a breeding surface but a protection surface, and the design accepts that trade explicitly.

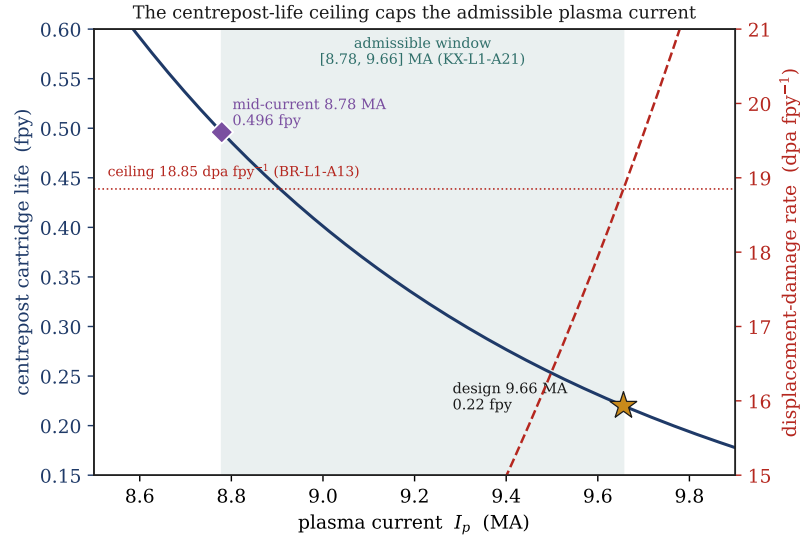
#### 4.5 The admissible current window

Because the fast fluence scales with source strength, the centrepost-life ceiling limits how hard the machine may be driven. Figure 7 shows the cartridge life and the displacement rate across the admissible current window of equation (13): at the design current 9.66 MA the life is 0.22 fpy against the  $18.85 \text{ fpy}^{-1}$  damage ceiling, and retreating to the mid-current edge 8.78 MA lowers the source strength enough to extend the life to 0.496 fpy ( $\times 2.25$ ; serial BR-SX-16). The curve is a phenomenological interpolation anchored to the two register operating points; the strong sensitivity reflects the combined reduction of plasma current and density at the mid-current point, not a



**Figure 6.** Coverage-versus-protection Pareto: net 1-D radial TBR against centrepost cartridge life (REBCO ceiling) as the WC/mantle split is swept (colour = WC thickness). Net TBR is below the 1.05 self-sufficiency line at every inboard split; self-sufficiency is met in the 3-D outboard closure. The corner design (star) gives all standoff to WC.

single-parameter power law. The current window is therefore the primary in-shot lever on cartridge life once the shield is fixed at the corner.



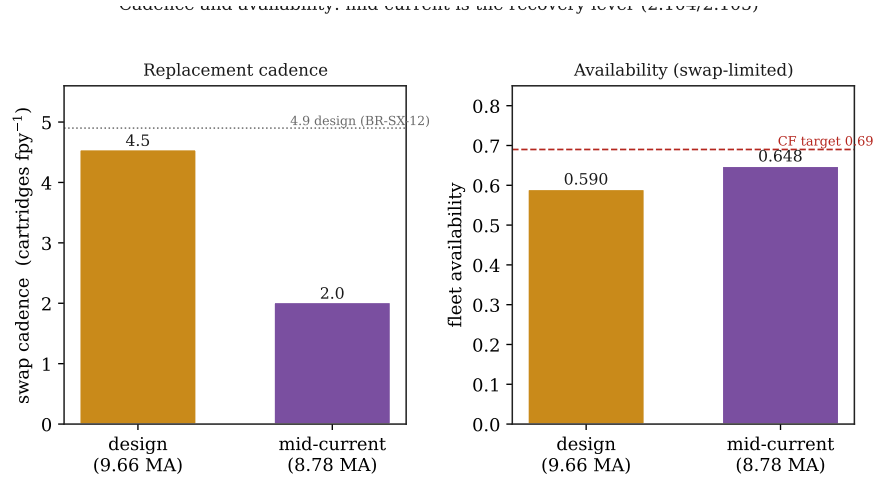
**Figure 7.** Cartridge life (navy, left) and displacement-damage rate (red, right) across the admissible plasma-current window [8.78, 9.66] MA (serial KX-L1-A21). The mid-current operating point extends life from 0.22 to 0.496 fpy; the dotted line is the 18.85 fpy<sup>-1</sup> damage ceiling.

#### 4.6 Replacement cadence and fleet availability

Equation (10) turns the design life directly into a replacement cadence:

$n_{\text{swap}} = 1/0.22 \approx 4.5$  swaps fpy<sup>-1</sup> at design (serial SC-BR-2; the 3-D CAD regression gives 4.9 swaps fpy<sup>-1</sup>, serial BR-SX-12), falling to  $\sim 2.0$  swaps fpy<sup>-1</sup> at mid-current. Figure 8 pairs the cadence with the swap-limited fleet availability carried in the RAMI companions: 0.590 at design and 0.648 at mid-current (serials BR-SX-18, BR-SX-16), against a target capacity factor of 0.69. The swap cadence pulls availability below the target at both operating points — an honest negative — and the mid-current point is the recovery lever, trading source strength (and hence isotope throughput) for uptime. The per-swap outage critical path ( $\sim 21.5$  d, dominated  $\sim 40\%$  by thermal cycling) and the  $\sim 39$ -swap thirty-year fleet history are analysed in the swap-logistics and

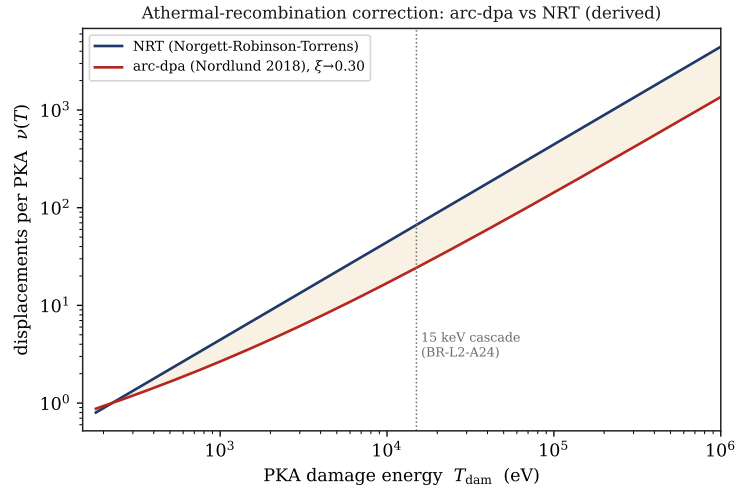
fleet-availability companions [40, 41]; the remaining-useful-life prognostics that schedule each swap are in the RUL companion [39].



**Figure 8.** Cartridge replacement cadence (left) and swap-limited fleet availability (right) at the design and mid-current operating points (serials BR-SX-12, BR-SX-18, BR-SX-16). Availability is below the 0.69 capacity-factor target at both points; mid-current is the recovery level.

#### 4.7 Displacement damage: NRT versus arc-dpa

Figure 9 shows the displacement function of equations (7)–(9): the NRT number grows linearly in damage energy, while the arc-dpa efficiency suppresses the surviving-defect count toward  $\xi_{\text{arc}} \rightarrow 0.3$  at the cascade energies ( $\sim 15$  keV) resolved by the MD study of serial BR-L2-A24. The register anchors the design-basis damage rate on the conservative NRT convention ( $18.85 \text{ fpy}^{-1}$ ), so the arc-dpa reduction is carried as unclaimed margin rather than as a life extension; reporting the NRT number keeps the cartridge cadence on the safe side of the true surviving-damage rate.



**Figure 9.** Displacements per PKA versus damage energy: NRT (equation 7) and the arc-dpa correction (equation 9,  $\xi \rightarrow 0.30$ ). (Derived model curve.) The design-basis  $18.85 \text{ fpy}^{-1}$  is reported on the conservative NRT convention; the shaded region is unclaimed athermal-recombination margin.

## 5 Discussion

### 5.1 Against the shielded-superconducting-core literature

The central finding — that a compact ST centrepost cannot be shielded to lifetime service within its inboard radius and is best operated as a consumable — agrees with, and quantifies, the shielded-core studies of Windsor *et al.*, who showed that even optimised tungsten- and tungsten-boride shields leave a superconducting centre column with a bounded, radius-limited

protected life [7, 8, 9]. Our effective removal coefficient of  $0.070\text{--}0.141\text{ cm}^{-1}$  for WC and the factor-169 attenuation over 45 cm are consistent with the order-of-magnitude-per- $\sim 20$  cm attenuation those studies report for high- $Z$  shields, and the depth-hardening of  $\Sigma_r^{\text{eff}}$  is the same buildup physics seen in point-kernel and discrete-ordinates shield calculations. The ARC demountable-magnet concept [3, 4] motivates the cartridge architecture directly: once the magnet is demountable for maintenance, a scheduled fluence-limited swap is a design choice rather than a failure mode.

### 5.2 The fluence ceiling in context

The cartridge life depends linearly on the assumed REBCO fast-fluence ceiling, so the central number (0.22 fpy) is only as firm as  $\Phi_{\text{allow}}$ . Our value  $3.9 \times 10^{22}\text{ n m}^{-2}$  sits squarely inside the measured critical-current-degradation onset for coated conductors under fast-neutron irradiation reported by Fischer, Prokopec and Eisterer [19, 20], and is set independently in the Kronos programme by first-principles DFT+ $U$  and cascade MD (companion [35]); the un-doped break-even fluence ( $5.64 \times 10^{22}\text{ n m}^{-2}$ ) and the  $1 \times 10^{23}\text{ n m}^{-2}$  optimistic bound bracket a factor of  $\sim 2.6$  in life. This is the dominant physics uncertainty on the cadence and is flagged as such.

### 5.3 Damage convention and structural life

Reporting  $18.85\text{ fpy}^{-1}$  on the NRT convention rather than  $\sim 5.7\text{ fpy}^{-1}$  on arc-dpa is a deliberate conservatism: the arc-dpa reduction (equation 9) is real, but folding it into the design rate would loosen the current window and the cadence, and we prefer the safe side. The first-wall and structural consequences of this damage rate — swelling, DBTT shift and helium embrittlement in the reduced-activation ferritic-martensitic steel and the tungsten first wall — are quantified in the companion first-wall studies [38, 25, 26] and are consistent with the fusion-materials limits [23, 24].

### 5.4 Benchmark synthesis

Table 4 places the centrepost against the two reference points most often invoked for compact, high-field devices: the ARC-class demountable centre column and generic ST fusion-nuclear-science-facility inboard builds. The Kronos centrepost is on the same side of the trade — radius-limited protection, consumable operation — with the added, quantified constraint that the inboard shield is also in competition with breeding.

**Table 4.** Benchmark synthesis. The compact ST centrepost is fluence-limited and radius-limited in the same regime as comparable high-field / spherical concepts; the distinguishing constraint here is the shield-versus-breeding competition on a shared inboard radius.

| attribute                  | Kronos centrepost (this work)        | comparable concepts                         |
|----------------------------|--------------------------------------|---------------------------------------------|
| inboard-shield attenuation | $\times 169$ over 45 cm WC           | $\sim$ decade / 20 cm high- $Z$ [8]         |
| protected life             | radius-limited, 0.22 fpy design      | radius-limited (ARC-class) [3]              |
| magnet strategy            | demountable, scheduled cartridge     | demountable magnets [3, 4]                  |
| REBCO fluence ceiling      | $3.9 \times 10^{22}\text{ n m}^{-2}$ | $10^{22}\text{--}10^{23}$ measured [19, 20] |
| distinguishing constraint  | shield vs. inboard breeding          | typically shield-only                       |

## 6 Limitations

One honest engineering gate remains open. The transport model is a 1-D radial infinite-cylinder CSG with reflective  $z$ -planes: it has no axial leakage, no divertor or port cutouts, and no toroidal-field ripple, and the triangularity  $\delta = -0.30$  is not represented, so its net TBR is an upper-ish estimate for a given radial build and the true 3-D coverage is lower (recovered in the CAD-faithful companion). Separately, the “life” uses fast-fluence damage-limit proxies (equation 10), not a fully spectrum-folded NRT-dpa failure model of the conductor, so the absolute cadence scales inversely with the chosen  $\Phi_{\text{allow}}$ . The single hard geometric gate is radial-budget closure: the centrepost winding plus a 30 cm WC shield occupies 0.74 m, which overruns the available  $R_0 - a = 0.72$  m inboard envelope by  $\sim 2$  cm (serials BR-SX-13, BR-SXD-06). Structural (181 MPa,  $6\times$  margin) and quench budgets already pass; the  $\sim 2$  cm overrun requires a shield/winding re-optimisation or a modest geometry relief and is carried as an explicit open item, not papered over.

## 7 Conclusion

The inboard centrepost is the defining vulnerability of a compact spherical-tokamak breeder, and this paper quantifies it end-to-end. Solving the fixed-source Boltzmann transport equation on

ENDF/B-VIII.0 with  $2.5 \times 10^9$  histories per point, the centrepost fast fluence attenuates by a factor of 169 over a 45 cm WC standoff, with an effective removal coefficient that hardens from 0.070 to  $0.141 \text{ cm}^{-1}$  as buildup is stripped away. Folded against the REBCO fast-fluence ceiling  $\Phi_{\text{allow}} \approx 3.9 \times 10^{22} \text{ n m}^{-2}$ , the design cartridge life is 0.22 fpy at a reference damage rate of  $18.85 \text{ fpy}^{-1}$ , tunable from 0.0048 (thin shield) to  $0.194 \text{ fpy}$  (thick shield) and extended to  $0.496 \text{ fpy}$  ( $\times 2.25$ ) at the mid-current edge of the admissible current window [8.78, 9.66] MA. The inboard shield trades directly against breeding (net 1-D TBR  $1.035 \rightarrow 0.923$ ), and the coverage-versus-protection optimisation resolves — by an explicit KKT argument — to the corner design: give the full standoff to shield and recover self-sufficiency from the outboard blanket. The design consequence is a maintenance architecture, a demountable scheduled-swap cartridge at  $\sim 4.5\text{--}4.9 \text{ swaps fpy}^{-1}$  with a swap-limited availability of 0.59–0.65, stated as such rather than wished away. Every headline quantity is a frozen anchor in the de-risking register, and the one open gate — a  $\sim 2 \text{ cm}$  inboard radial-budget overrun — is named, not hidden.

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**Reproducibility.** Every headline quantity is a frozen anchor in the Kronos Physics De-Risking Register [33] (doi:10.5281/zenodo.22133057): the shield trade and fast-fluence attenuation (BR-L1-A13b), the reference damage rate (BR-L1-A13), the REBCO fast-fluence ceiling (BR-SX-04), the operating-point cartridge-life anchors (BR-SX-16, SC-BR-2) and the current window (KX-L1-A21). The OpenMC 0.15.3 model, the ENDF/B-VIII.0 inputs, the radial build and the figure-generation script are archived with this paper at doi:10.5281/zenodo.22645708; official version:

<https://www.kronosfusionenergy.com/publications/centrepost-fluence-and-shielding.html>; no result depends on unpublished or proprietary data, and no economic quantity appears in the deposit.

### Data availability

The shield-trade tallies, the fast-fluence and net-TBR tables, the removal-coefficient and cartridge-life reductions and the figure scripts used for figures 3–9 are archived with the deposit at doi:10.5281/zenodo.22645708. All quantitative values trace to the register [33] by the serial identifiers cited inline.

### Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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